# Uncovering Dynamic Consumer Behavior Patterns Across Commercial Spaces: A Mobility-Based Evidence of Core Consumer Groups in the 23 Wards of Tokyo

**Abstract**

As consumption demands evolve, consumer schedules and mobility have become increasingly fragmented, shifting commercial competition from physical scale expansion toward alignment with daily mobility contexts. However, conventional static survey methods often fail to capture the dynamic nature of consumer activities, hindering evidence-based commercial development. This study addresses this gap by developing a mobility-based framework that integrates temporal, spatial, and activity-sequence dimensions. Using the 23 wards of Tokyo as a case study, we utilize high-resolution GPS data and a multi-stage filtering process to identify consumers across five facility categories. By introducing a Facility Choice Preference Index (FCPI), we identify five core consumer groups with shopping preference intensity: prime-aged women, prime-aged men, young women, senior women, and adolescent females. Results indicate that shopping activities predominantly peak in the afternoon, characterized by short durations ($\leq$60 minutes) and localized spatial ranges ($\leq$0.5 km). These activities are primarily governed by walking and neighborhood scales, with regional-scale movements playing a secondary role. Furthermore, shopping exhibits strong sequence continuity, frequently anchored by home and obligatory activities. We further identify four consumer behavior patterns—routine, work-embedded, neighborhood, and school-anchored shopping—illustrating how commercial spaces are integrated into consumers' daily activity chains. These findings offer micro-scale insights for optimizing commercial layouts and urban support systems, ultimately enhancing the vitality and resilience of urban commercial ecosystems.

*Keywords:* core consumer group, shopping activities, commercial facilities, activity sequence, mobility data

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## 1. Introduction

Urban economic vitality is a key foundation for urban economic resilience and sustainable development[? ]. As spatial carriers of the urban economy, commercial spaces are not merely sites of static transactions but act as dynamic nodes that connect production, everyday life and urban mobility. They facilitate essential consumption, social interaction and the spatiotemporal linkage of diverse activities[? ]. Ongoing urbanization, economic growth, and demographic change have reshaped consumption demand and expectations, making consumer schedules and mobility patterns increasingly fragmented and diverse[? ? ]. At the same time, even though the growth of e-commerce has intensified competition, many consumers still desire experiences offered by physical commercial spaces[? ]. Consequently, the focus of commercial competition is pivoting from spatial scale expansion toward behavioral alignment[? ]. In this evolving context, urban commercial facilities must be reconceptualized: they are no longer passive containers of demand, but integrative functional nodes deeply embedded within the continuity of consumers' daily activity chains.

However, prevailing commercial evaluation, planning strategies, and consumer behavior research still rely primarily on static indicators such as aggregate visitor volume, density, and physical accessibility[? ? ? ]. While these metrics are useful for describing facility distribution and baseline service capacity, they fundamentally overlook the continuous mobility flows that embed shopping within consumers' daily routines[? ]. This reliance on static observation creates critical knowledge gaps in three interconnected dimensions. First, it obscures the micro-level behavioral mechanisms underlying commercial resilience[? ]. Commercial facilities with similar visitor volumes may depend on entirely different behavioral logics—such as commute-embedded consumption versus purpose-driven trips. When social rhythms shift (e.g., the normalization of remote work), facilities weakly anchored in daily sequences become highly vulnerable to demand shocks, a dynamic that static data cannot predict. Second, a purely distance-based view of accessibility masks schedule-dependent inequalities[? ]. Physical proximity fails to capture the actual time costs faced by different demographic groups, as it ignores the rigid constraints of their pre- and post-shopping activity sequences. Third, aggregate footfall data inevitably conflates active and preference-driven con-



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sumers with passive passersby. This dilution makes it exceptionally difficult to identify the true usage contexts of a facility. Compounding these methodological limitations is a pervasive issue of scope: existing studies frequently confine their analysis to a single facility category or report only generalized, city-wide patterns[? ? ? ? ]. This fragmented approach severely limits cross-type comparisons, leaving a blind spot regarding how consumer behaviors systematically differ across diverse commercial spaces. Consequently, to enable meaningful analysis and robust commercial planning, there is a pressing need to shift from an absolute-volume perspective to a preference-intensity perspective, which can accurately isolate core consumer groups and decode their spatiotemporal activity sequences.

Currently, the growing availability of digital trajectory data provides a strong empirical basis for examining these mechanisms and reshaping consumer-behavior research[? ]. Compared with traditional travel surveys, which are often constrained by high implementation costs, limited sample size, recall biases, low temporal granularity and narrow commercial categorical coverage[? ? ? ? ? ? ? ], survey-based approaches can yield a static and partial view that misses high-frequency and spatially complex behaviors[? ? ]. These high-resolution datasets enable continuous observation of consumer spatiotemporal trajectories at city scale and with greater objectivity[? ? ? ? ]. However, a critical gap often remains between the reliable extraction of mobility events from noisy, anonymous traces and their behavioral interpretation. In particular, the robust identification of behavioral semantics and external validation at the facility level remain persistently limited[? ? ? ? ? ? ? ].

Consequently, this paper leverages high-resolution GPS trajectory data within the 23 wards of Tokyo to bridge the gap between raw mobility events and behaviorally meaningful insights. To prevent behavioral patterns from being diluted by incidental or passive visits, we first utilize a four-step filter and then introduce a Facility Choice Preference Index (FCPI) to extract core consumer groups with a high intensity of shopping preference. By integrating temporal, spatial, and activity-sequence dimensions, this research expands the conventional static analytical perspective into a dynamic framework that captures consumer behavior patterns. This study is guided by three core research questions: 1) Who are core consumer groups? 2) What are the temporal, spatial and activity-sequence characteristics of consumer behavior? 3) How do consumer behavior patterns differ across commercial spaces and day types to inform resilient commercial development? This study focuses



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on urban commercial spaces in the 23 wards of Tokyo, where we classify the diverse commercial spaces into five analytical categories: commercial-type (SCC), office-type (SCO), train station-type (SCT), and residential-type (SCR) shopping centers, alongside shopping streets (SS).

## 2. Related work

### *2.1. Consumer behavior research*

Much of the existing consumer behavior research has predominantly relied on aggregate visitor volumes and static spatial accessibility, which can be broadly categorized into two approaches. The first approach evaluates retail performance through absolute traffic data. For instance, Philp et al. utilize daily and weekly footfall to understand retail behaviors[**?** ], while Graham et al. explore the relationship between traffic and retail sales[**?** ]. However, analyzing consumer behavior solely through the lens of total visitor volumes presents significant methodological drawbacks. Full visitor volume inevitably captures many random passersby without any genuine shopping preference, which disturbs the precise identification of core consumer characteristics. Furthermore, relying on such aggregate metrics can introduce demographic-structure bias[**?** ]. The second approach emphasizes spatial proximity. Xu et al. examine grocery-store visitation in relation to travel distance and neighborhood proportions[**?** ], and Shao et al. analyze the effects of calculated accessibility to metro stations on shopping travel flexibility[**?** ]. While these static spatial indicators provide baseline geographical insights, they often underestimate inequality in consumption opportunities. As highlighted by Niedzielski et al.[**?** ], physical proximity alone does not reflect the schedule-dependent time costs that vary significantly across different population groups. Consequently, the reliance on these isolated static indicators—whether volume or distance—obscures the continuous activity flows that link shopping with other daily routines. This methodological gap makes it difficult to capture facility-specific consumer–place relationships and the underlying behavioral logics essential for commercial resilience in a dynamic environment[**?** ].

Parallel to urban studies, consumer behavior literature within the marketing domain has heavily prioritized psychological and promotional factors[**?** **?** ]. For example, extensive research has explored the effects of online consumer reviews[**?** ], social marketing strategies[**?** ], and sensory stimuli[**?** ].



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However, these perspectives typically operate in a spatial vacuum. They frequently overlook the crucial reality that consumer choices are deeply embedded within spatiotemporal contexts and daily mobility routines. As Birkin et al.[**?** ] and Rozenkowska[**?** ] argue, consumer behavior is inherently dynamic and shaped by a multitude of interdependent spatial constraints. By ignoring the physical trajectory of consumers, these marketing-centric approaches lack the explanatory power to fully unpack urban commercial resilience.

Furthermore, even when spatial dimensions are considered, prevailing approaches frequently suffer from oversimplification in both facility scale and activity categorization. On one hand, many empirical studies confine their scope to a single type of commercial facility, such as isolated shopping centers[**?** ] or specific shopping streets[**?** ]. This fragmentation severely constrains the generalization of their findings across a diverse urban commercial network. Similarly, classifying shopping spaces primarily by static built-environment attributes, such as size and store counts[**?** ], fails to capture the heterogeneous and evolving functions of contemporary facilities. On the other hand, traditional travel models often treat shopping as a monolithic, undifferentiated activity category[**?** ]. This reductionist approach obscures the complex, multipurpose nature of modern consumer trips, highlighting a critical need for a holistic framework that examines diverse commercial facilities through the lens of continuous activity sequences.

To address these critical gaps, this study proposes a mobility-based framework that integrates temporal, spatial, and activity-sequence dimensions. By capturing the continuous trajectories of diverse consumer groups, this approach aims to unpack the underlying behavior patterns and provide data-driven insights for resilient urban commercial planning.

### *2.2. Research on consumer behavior from the perspective of mobility*

Traditional research on consumer behavior has often relied on personal trip surveys and questionnaire surveys to collect travel information[**? ? ? ?** ]. While informative, these methods capture limited detail and sample size, and are vulnerable to recall and sampling biases[**? ? ? ? ?** ]. More importantly, this reliance on self-reported, temporally static data creates a fundamental bottleneck for understanding the continuous and dynamic complexity of modern consumer behavior.

The analysis of human mobility and behavior patterns dates back to the late 1950s [**?** ]. Advances in ubiquitous mobile sensing and spatial data infrastructure have turned big data into a practical remedy for observing consumer



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behavior at scale. The widespread adoption of GPS-enabled devices provides high-resolution, continuous trajectories of human mobility, effectively overcoming the sample size and temporal limitations of traditional surveys[? ? ? ? ]. Consequently, mobility-based analytics have been increasingly adopted to decode dynamic behavior patterns in urban and commercial contexts, transforming how researchers conceptualize spatial-temporal consumption[? ? ? ? ? ? ].

However, studying consumer behavior in commercial facilities via raw mobility data remains methodologically challenging. First, raw GPS traces lack semantic context. It is profoundly difficult to distinguish genuine shopping activities from non-commercial stops or mere passing by[? ? ? ? ]. Furthermore, because these large-scale datasets are rigorously anonymous to protect privacy, linking trajectories to individual sociodemographic profiles or specific commercial intents is highly problematic. Second, validating these identified consumer behaviors is persistently hindered by the scarcity of fine-grained, facility-level ground-truth business data[? ]. As previous studies demonstrate, researchers often resort to indirect validation. For example, Takahashi et al.[? ] validated estimates against turnover, assuming comparable temporal co-movement even when absolute magnitudes diverged. Similarly, Li et al.[? ] relied mainly on visualization and plausibility checks without quantitative accuracy metrics.

Collectively, these limitations underscore an urgent need for methodological frameworks in raw data preprocessing and behavior identification. Specifically, overcoming the anonymity and noise of GPS data requires advanced approaches to (1) extract reliable mobility events (e.g., stays and visits), (2) distinguish actual consumers from passers-by, office workers, and facility staff, and (3) decode continuous behavioral characteristics with explicit external validation[? ? ? ].

## 3. Methodology

### *3.1. Study area and data source*

This study is situated in the 23 wards of Tokyo, the core of the world's largest metropolitan area (Figure 1), with a population of 10 million people and an area of 627 $km^2$. As of 2022, tourism- and recreation-related expenditures in Tokyo Prefecture totaled 5,090.8 billion yen[? ], and department stores and supermarkets in the 23 wards recorded annual merchandise sales of 483,207 million yen with a total sales floor area of 634,000 $m^2$[? ]. In the



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same year, the 23 wards contained 228 shopping centers (SCs)[? ] and 783 shopping streets (SSs)[? ], representing 5.21% and 7.02% of Japanese totals.

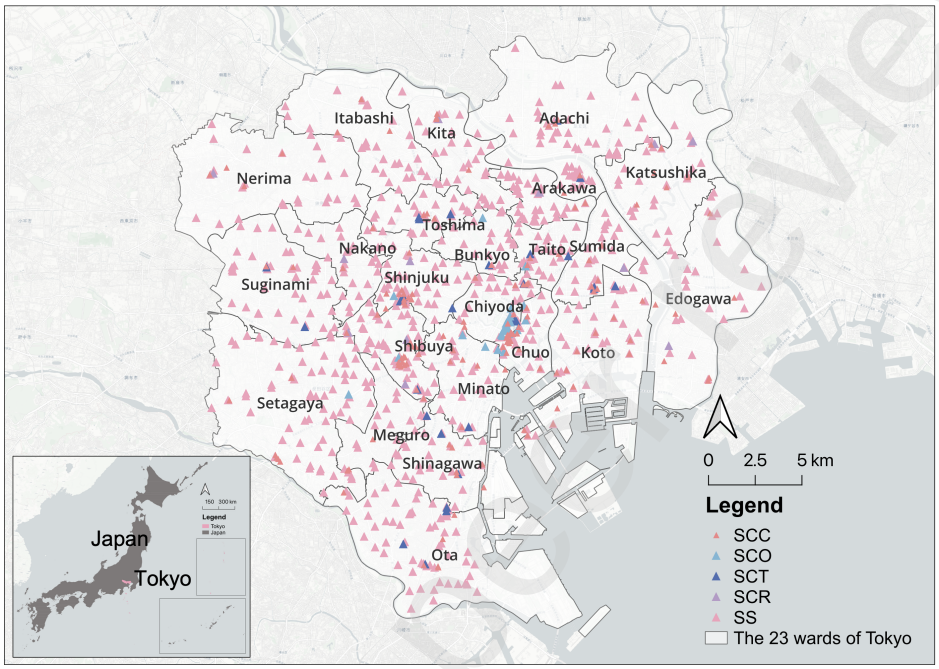



Figure 1: Study area and the spatial distribution of commercial facilities

*Note:* SCC = commercial-type shopping center; SCO = office-type shopping center; SCT = train station-type shopping center; SCR = residential-type shopping center; SS = shopping street.

Three datasets are utilized in this study:

*Mobility data*

We used GPS point-type data from the 23 wards of Tokyo, provided by AGOOP Company. This dataset was generated from a multitude of smartphone applications, and it had 1.135 million monthly users in Tokyo. From the continuous data stream, we selected mobility data from a one-month period for our analysis from May 1 to May 31 of 2022, comprising nineteen weekdays, twelve weekends and holidays. A primary strength of this dataset is its capacity to trace individual behavior patterns within a 24-hour period. Each location log is a rich data point that includes age, gender, family information, timestamp (down to the second), spatial information (latitude and longitude), home city code, workplace city code, etc. We preprocessed the



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data by removing duplicate records, flagging observations with missing values in *gender*, *age*, *household income* and *marital status*, and filtering individuals whose *home city codes* corresponded to the 23 wards of Tokyo. To correct for sample size bias in GPS data and align the sample with the population structure of the 23 wards of Tokyo, we compared the GPS-derived population against the 2020 National Census population at the chōme (a Japanese sub-district numbering system) level [**?** ], and computed magnification factors for each chōme. After this preprocessing, we got an GPS trajectory dataset comprised 497,921,842 GPS trip records in thirty-one days.

*Commercial dataset*

We utilized a shopping center point dataset provided by the Japan Council of Shopping Centers (JCSC), which documents all SCs in the 23 wards of Tokyo established before 2023. Each point in this dataset represents an individual SC and contains attributes such as its official name, building type, spatial information (latitude and longitude), etc. Concurrently, we utilized a polygonal shopping street dataset compiled by Garcia et al [**?** ], which delineated the spatial boundaries of individual SS across the 23 wards of Tokyo.

*Building Footprint data*

We integrated vector polygonal dataset [**?** ], which provides building footprints within the 23 wards of Tokyo in 2022.

*POI dataset*

We utilized a point-of-interest (POI) dataset [**?** **?** ], which provides the geographic locations and categorical information of urban facilities within the 23 wards of Tokyo. It covers various types of facilities: public service facilities, such as school, hospital government institutions, post offices, welfare facilities, government offices, police stations, post offices, fire stations and cultural facilities; transport facilities, such as bus stations, rail stations and airports, public service; shopping facilities, such as convenience stores.

*Land use dataset*

We utilized a land use dataset [**?** ], which provides the geographic locations and categorical information of land-use classes within the 23 wards of Tokyo. It includes forest, parks and green spaces, rivers and lakes, beaches and other land-use types.

### *3.2. Analytical framework*

Unlike routine commuting, shopping activities exhibit a high degree of spatiotemporal flexibility. Given the diversity and abundance of urban com-



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mercial spaces, consumers demonstrate substantial mobility in their facility choices. Rather than being isolated activities, shopping activities are dynamic processes embedded within broader daily activity chains, such as home-, work-, and school-related routines.

Accordingly, moving beyond traditional static perspectives, this study adopts a mobility-based four-step analytical framework, as illustrated in Figure 2: First, mobility data preprocessing establishes the analytical foundation through raw GPS data cleaning, attribute flagging, spatial filtering and the application of population expansion factors (detailed in Section 3.1). Second, spatial-temporal definition of commercial space delineates commercial facility categories, the spatial boundaries and effective business hours. Third, identification of core consumer groups is executed in two stages: (1) identifying consumers from GPS trajectories dataset by filtering out passers-by, office workers and facility staff; and (2) identifying core consumer groups with shopping preferences by utilizing the Facility Choice Preference Index (FCPI) across consumer attributes and diverse facility categories. Fourth, the analysis of consumer behavior patterns compares the temporal, spatial, and activity-sequence characteristics of these groups across weekdays and non-weekdays to synthesize representative consumer behavior patterns. Ultimately, this framework provides evidence-based insights for the micro-level optimization of commercial spaces, supporting the long-term vitality and resilience of urban commercial systems.

### *3.3. Spatial-temporal definition of commercial space*

#### *3.3.1. Commercial space classification*

Japanese business policy, particularly the Large-Scale Retail Store Law, has long structured Japanese commercial landscape around a core conflict between large-format and small-scale stores[**?** ]. This policy-driven history resulted in two distinct types of commercial environments-Shopping Center (SC)-which act as the primary vehicle for large-format and corporate retail, and Shopping Street (SS), the spatial embodiment of small-scale local commerce. The fundamental differences between these two forms directly influence consumer behavior, and a clear definition of each is therefore the essential first step for a comparative analysis. To process this analysis, this study defines and classifies "commercial facilities"based on the following principles.

SCs are planned commercial developments, typically developed and managed by large private enterprises[**?** ]. Characterized by unified, large-scale



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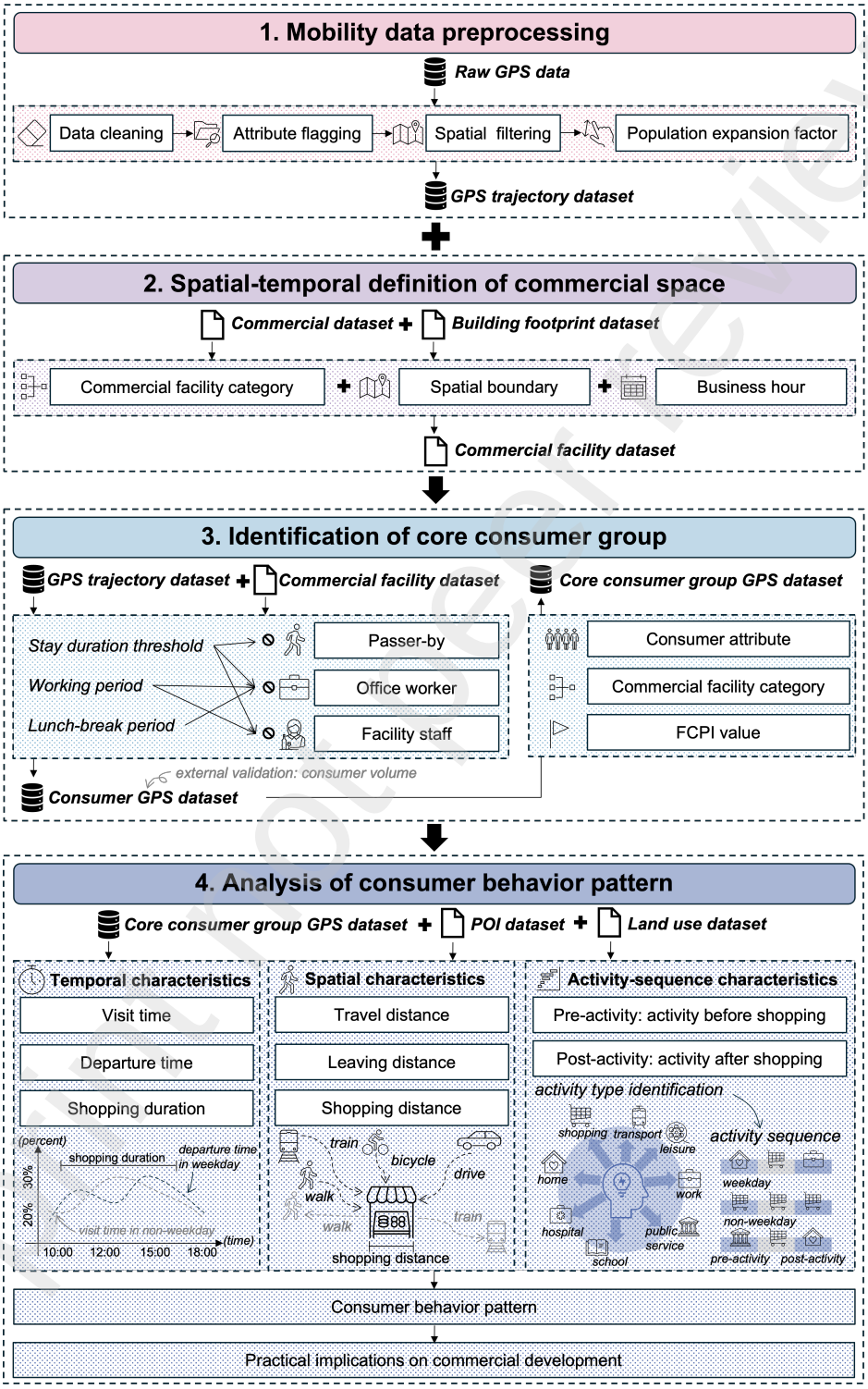



Figure 2: Analytical framework



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buildings with curated tenant mixes and extensive parking infrastructure, they create a controlled spatial environment. This design encourages prolonged, purpose-driven consumer visits and structured shopping experiences, with a retail composition dominated by major brands and chain stores. In contrast, SSs have evolved organically from the historical agglomeration of independent small retailers[**?** ]. Beyond their commercial role, these streets are integral to urban vitality and community cohesion[**?** ]. Their open and street-level environment, populated by numerous small-scale and specialized shops, fosters exploratory, spontaneous, and leisurely consumer behavior. This diverse composition promotes social interaction and discovery rather than purely transactional and destination-focused shopping.

While SSs form a single analytical category, the SC category required further refinement. The initial JCSC dataset classified SCs into nine building types. To create a more analytically robust typology, we refined this classification into four primary groups through exclusion and consolidation. First, three categories were excluded: airport terminals, as they are specialized travel-dependent destinations; hotels, due to significant data incompleteness; and the single underground building, which was removed as a statistical outlier. Next, we consolidated the commercial, sub-viaduct, and mixed-use building types into a unified "commercial" category, justified by their shared primary function as destinations for public commercial activity. This refinement process resulted in our four final analytical categories for SCs: commercial-type shopping center (SCC), office-type shopping center (SCO), train station-type shopping center (SCT), and residential-type shopping center (SCR). These four SC categories, together with SS, form the five distinct facility types examined in this study, as illustrated in Figure. 3.

Therefore, in this study, we constructed a commercial facility dataset, covering information of spatial boundary and business hour of SCs and SSs within the 23 wards of Tokyo.

*3.3.2. Spatial boundary definition*

**Shopping center**

Regarding the identification of shopping center boundaries, we utilized shopping center point dataset overlaid by the building footprint data. Then 212 SCs were retained after excluding three airport terminals, three hotels, and one underground building due to specialization or data incompleteness (detailed in Section 3.3). Then, a 30-meter buffer was applied individually to each shopping center footprint to define the shopping activity area, allowing



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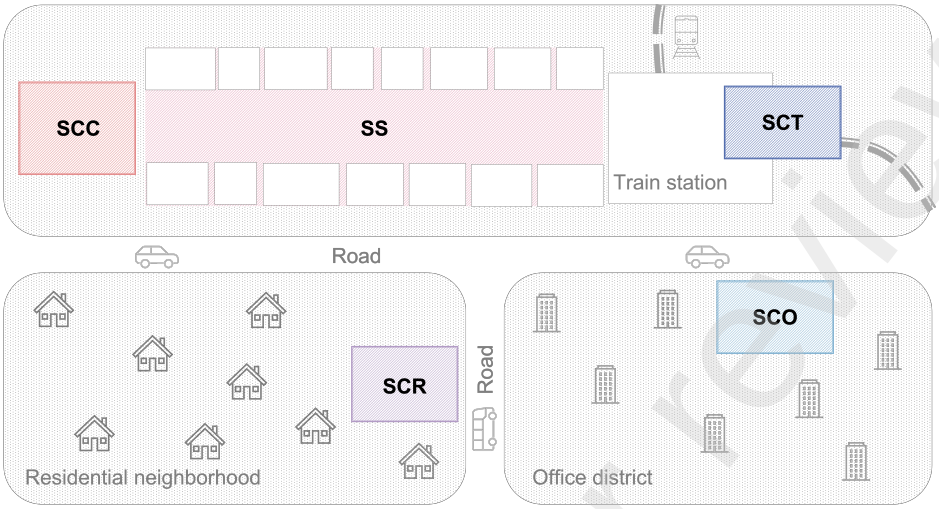



Figure 3: Illustration of five commercial facility categories

*Note:* SCC = commercial-type shopping center; SCO = office-type shopping center; SCT = train station-type shopping center; SCR = residential-type shopping center; SS = shopping street.

coverage of sidewalk edges and entrance plazas while minimizing overlap with surrounding buildings. Fig. 1 shows the spatial distribution of commercial facility boundaries across the 23 wards of Tokyo.

**Shopping street**

The boundaries of SSs within the 23 wards of Tokyo were identified using an agglomeration clustering method from Garcia et al [**?** ] based on the methods proposed by MacDonald et al. [**?** ] and Ballantyne et al. [**?** ], both of which utilize Uber's H3 hexagonal grid system. First, telephone directory data[**?** ] is implement and a threshold adjustment is processed suitable for high retail density in the 23 wards of Tokyo. Cells with two or fewer businesses were excluded. Subsequently, shopping street clusters were formed by grouping contiguous cells with sufficient business presence and their adjacent connectors. Then, SS clusters with fewer than 30 stores were excluded from further analysis to reduce noise, particularly in central districts. Finally, the resulting SS clusters were matched to a manual placement.

*3.3.3. Business hour definition*

Based on common business patterns across the 23 wards of Tokyo, we defined two sets of hours: for SCs, the business hours were set from 10:00 to



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20:00; for SSs, which often feature establishments with later hours such as Japanese pubs and restaurants, we defined broader business hours from 10:00 to 24:00. And they were used in the identification of consumer (detailed in Section 3.4.1) to restrict valid shopping visits.

### *3.4. FCPI-based identification of core consumer groups*

#### *3.4.1. Identification of consumer from GPS trajectory dataset*

In this study, consumers are defined as individuals whose GPS trajectory intersect with the spatial boundaries of commercial facilities. A four-step filtering procedure was applied to identify valid consumers from the GPS trajectory dataset.

First, we applied a stay point detection algorithm implemented in the Scikit-mobility toolkit [**?** ] to construct GPS stay data. This method identifies stationary behavior based on two key parameters: a spatial distance threshold and a minimum time duration. We set the spatial threshold to 200 meters. To establish a suitable stay duration threshold for distinguishing consumers from passers-by, we chose twelve commercial facilities with available pedestrian volumes and conducted a sensitivity analysis across candidate stay durations (1, 2, 5, 10, 15, and 20 minutes). Giving the diminishing changes beyond 10-minute threshold and the stable estimated volumes at 15-minute threshold in Figure 4, together with a good fit ($R^2$ is 0.55 at 15-minute threshold) in Figure 5, we adopt a 15-minute stay duration threshold for the subsequent analyses presented in this paper. Therefore, we got a 15-minute GPS stay dataset comprised 18,807,339 GPS records with 12,006,844 weekday records, 6,800,495 non-weekday records. Also, we validated the consumer volumes from 15-minute stay data against real pedestrian volumes.

Second, to define valid shopping activity time intervals, and to account for positioning errors in GPS data, we treated stay-point records with arrival times up to 30 minutes before opening hours and departure times up to one hour after closing hours as valid shopping activity intervals.

Third, to filter out office workers, we defined a primary working period from 7:00 and 17:00 based on commuting patterns identified in the sixth Tokyo Metropolitan Area Person Trip Survey (2021), and a lunch-break period from 12:00 and 13:00, allowing a 15-minute tolerance to account for positioning errors. Individuals whose cumulative stay duration within commercial facilities exceeded seven-hour during the working period (with an additional one-hour tolerance for positioning errors) were classified as office workers. And their stay records observed during the working period were



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excluded from the analysis, except those occurring within the lunch-break period.

Fourth, to distinguish facility staff from consumers, we applied a three-hour stay duration threshold. This threshold was informed by two data references: the national average daily duration for part-time work is approximately 2.79 hours (Labor Force Survey, 2022)[**?** ], and the stay duration of typical consumers in Tokyo rarely exceed one hour [**?** ]. Accordingly, individuals with more than four-hour cumulative stay duration at the same commercial facility were classified as facility staff and were excluded from the analysis.

Therefore, we got a consumer GPS data comprised 5,987,727 consumer trip records with 3,444,920 weekday trip records, 1,935,798 weekend trip records, and 607,009 holiday trip records.

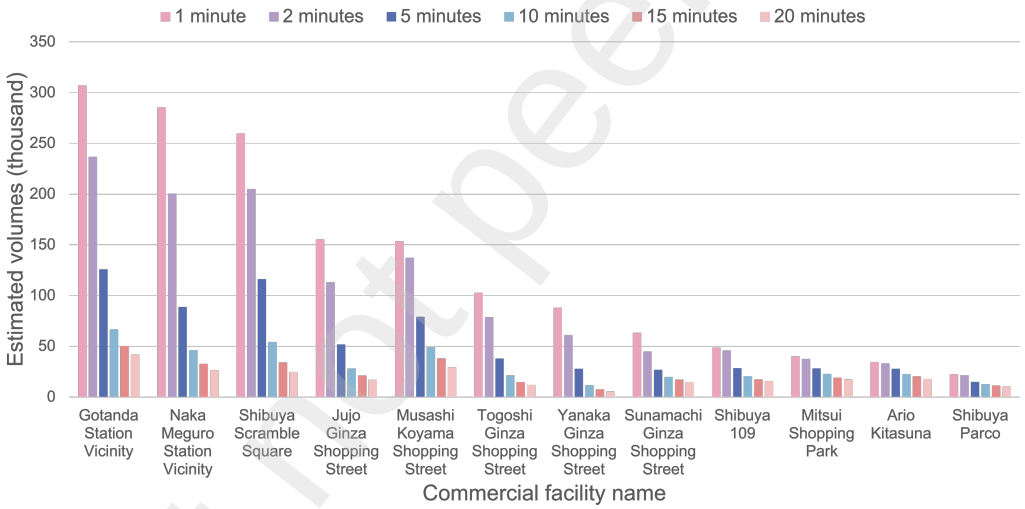



Figure 4: Estimated volumes against pedestrian volumes

### *3.4.2. FCPI-based identification of core consumer groups from consumer GPS data*

Unlike conventional approaches that identify consumers by absolute consumer volume, we analyze consumer behavior using *core consumer groups*, which exhibit high preference intensity after normalizing for demographic size. This choice is essential to capture the active vitality of commercial space rather than its passive footfall scale. We apply this concept using the



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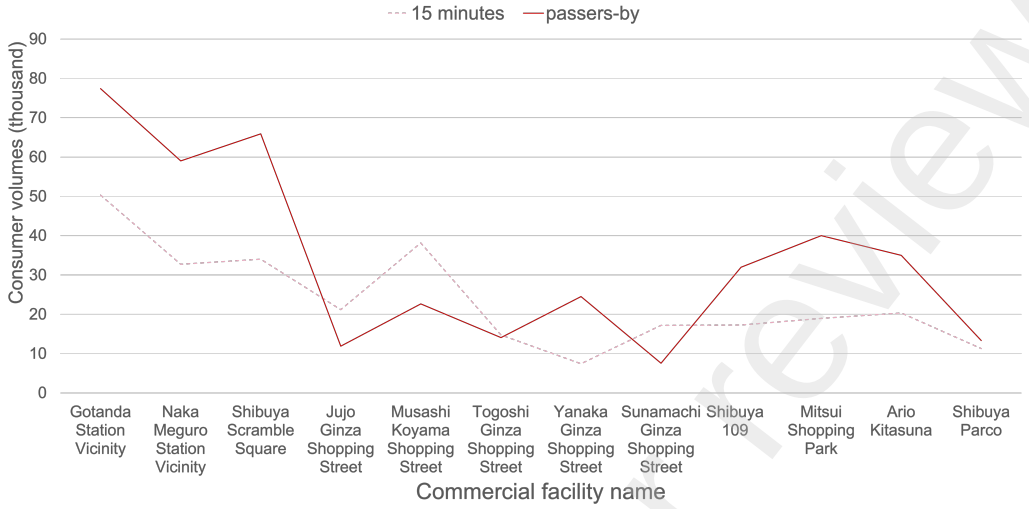



Figure 5: Sensitivity analysis of 15-minute stay duration threshold

Facility Choice Preference Index (FCPI), which measures the relative preferences intensity of consumer for facility category to its baseline prevalence in the full GPS data. By focusing on core consumer groups, the subsequent analyses are conducted on consumer groups who actively select a given facility category, thereby improving interpretability and enabling meaningful comparisons across categories. For example, prime-aged adults (30-49) may contribute to the largest proportion of total consumer volume because of their larger population base, while young adults (20-29 years) may exhibit stronger shopping preference once group size is taken into account.

*Consumer group attributes*

We processed *age*, *gender*, *marital status* and *household income* of consumers to express consumer demographic composition directly. Based on the population structure of census, we divided *age* into five types: under 19 years (adolescents), 20-29 years (young adults), 30-49 years (prime-aged adults), 50-69 years (senior adults), and over 70 years (older adults), divided *gender* into male and female, divided *marital status* into single person and married person, and divided *household income* into under 300 million yen (low household income), 300 million to 500 million yen (middle household income), 500 million to 700 million yen (middle-high household income) and over 700 million yen (high household income). In this study, we define *age* and *gender* as the primary attributes, while *marital status* and *household income* serve



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as secondary attributes for descriptive analysis.

*Facility Choice Preference Index (FCPI)*

The FCPI is computed in two steps. First, we calculate the baseline proportion of consumer group $c$ in the full consumer volume and its within-category proportion in facility category $f$ to its baseline proportion in the consumer GPS data (Equation 1 and Equation 2). Second, we compute the ratio of these two proportions to obtain FCPI (Equation 3). A FCPI value greater than 1 indicates relatively high preference, a value below 1 indicates relatively low preference, and a value equal to 1 indicates parity with the baseline. Consumer group aged over 70 were excluded from FCPI estimation because sparse observations can produce unstable ratio estimates and inflate FCPI variance.

$$P_{GPS}^{(c)} = \frac{N_{GPS}^{(c)}}{N_{GPS}} \tag{1}$$

where $N_{\text{GPS}}^{(c)}$ is consumer volumes in group $c$ in the full consumer GPS data, and $N_{\text{GPS}}$ is full consumer volumes in 15-minute stay data.

$$P_f^{(c)} = \frac{N_f^{(c)}}{N_f} \tag{2}$$

where $N_f^{(c)}$ is consumer volumes of group $c$ who visited commercial facility category $f$, and $N_f$ is full consumer volumes who visited commercial facility category $f$.

$$\text{FCPI}_f^{(c)} = \frac{P_f^{(c)}}{P_{GPS}^{(c)}} \tag{3}$$

where $\text{FCPI}_f^{(c)}$ denotes the FCPI value for consumer group $c$ in facility category $f$.

*Core consumer groups*

We identified core consumer groups $k$ within each facility category $f$ in three steps. First, we define preferred groups by applying three filters to each consumer group $c$, to ensure interpretability and exclude small or sparse groups,: (1) $FCPI_f^{(c)} \geq 1.2$; (2) the weighted consumer volume satisfies $N_f^{(c)} \geq 1000$; and (3) the within-category proportion satisfies $S_f^{(c)} \geq 0.5\%$. Second, for each preferred group, we compute a lift score (Equation 4) and its proportion among preferred groups $P_f^{(c)}$, and combine them into a preferred-group score (Equation 5). Third, within each facility category $f$, we rank



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preferred groups by preferred-group score and select the top five as the core consumer groups $k$.

$$LS_f^{(c)} = \frac{\overline{FCPI_f^{(c)}}}{1.2} \tag{4}$$

$$PGS_f^{(c)} = P_f^{(c)} \times LS_f^{(c)} \tag{5}$$

where $\overline{FCPI_f^{(c)}}$ is the aggregated FCPI of consumer group $c$ within facility category $f$; $LS_f^{(c)}$ measures preference intensity relative to the cutoff (1.2); and $PGS_f^{(c)}$ is the preferred-group score that jointly captures preference intensity and the group's contribution among preferred groups.

### *3.5. Analysis of consumer behavior patterns through temporal, spatial and activity sequence dimensions*

In this section, we analyze temporal, spatial and activity-sequence patterns of core consumer groups in Section 3.4.2 to explore consumer behavior patterns, separately for weekdays and non-weekdays.

#### *3.5.1. Temporal patterns*

In this study, temporal patterns are characterized using three indicators: visit time, departure time and shopping duration. Visit time is defined as the arrival time at a commercial facility category. Departure time is defined as the departure time after shopping within the same facility. Shopping duration is measured as the cumulative time between the first arrival and the last departure within the same facility on the same day. We define temporal context into four periods by time of day based on the business-hour definition in Section 3.3.3: morning (10:00–11:00), noon (11:00–14:00), afternoon (14:00–18:00), and night (18:00–24:00).

#### *3.5.2. Spatial patterns*

In this study, spatial patterns were represented as the travel distance, shopping distance and leaving distance. For each consumer, travel distance is calculated using the haversine from pre-activity location to shopping activity location. Their cumulative shopping distance was calculated as the shopping distance. Leaving distance is defined as the travel distance after shopping within the same facility. We define spatial context into four scales: walking scale (less than 0.5 km), neighborhood scale (0.5-2 km), regional scale (2-18 km) and metropolitan scale (more than 18 km).



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*3.5.3. Activity-sequence patterns*

We examine activity-sequence patterns of core consumer groups to reveal how shopping activities are embedded in consumers' daily routines and how they connect to surrounding activities. In this study, an activity-sequence is defined as a three-step chain centered on a shopping activity, consisting of the activity before shopping (pre-activity), the shopping activity itself, and the activity immediately after shopping (post-activity).

Our analysis uses two datasets with different roles. The core consumer group GPS dataset in Section 3.4.2 contains trajectory records of shopping activities made by consumers in the identified core consumer groups, restricted to traces observed within commercial facilities. To construct pre- and post-activities, we rely on the 15-minute stay dataset in Section 3.4.1, which records consumers' consecutive stay activities throughout the day. For each shopping activity, we define the pre-activity and post-activity as the immediately preceding and following stay activities in the 15-minute stay dataset. We then assign activity types to these pre- and post-activities by integrating spatial–temporal features, including inferred home and workplace locations and surrounding POIs and land-use context. The following subsections describe the identification of home and workplace locations and the activity types in detail.

*Identification of home and workplace locations*

Following [**?** ], we identify each consumer's home and workplace using a three-step procedure. First, we aggregate each consumer's stay locations into spatial clusters within a 200 m radius. To ensure behavioral regularity, we retain only clusters visited on at least 50% of the consumer's observed days. The retained clusters are ranked by the number of covered days, with ties broken by the total accumulated stay duration. These retained clusters define the candidate "activity locations" for subsequent home and work assignment. Second, we label the candidate clusters as home or workplace by matching their administrative city codes to each consumer's recorded home and workplace city codes. We consider four scenarios: (1) the home and workplace city codes are the same and only one cluster is available; (2) the home and workplace city codes are the same but multiple clusters are available; (3) the home and workplace city codes differ and only one cluster is available; and (4) the home and workplace city codes differ and multiple clusters are available. In each scenario, we select the cluster(s) whose city code(s) are consistent with the corresponding recorded code(s), prioritizing



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clusters with more covered days and longer accumulated stay duration. The centroid of the selected cluster is taken as the coordinate of home and workplace. Third, if home or workplace cannot be assigned in step two, we apply a time-based auxiliary rule. For home identification, we define nighttime as 21:00–08:00 and select candidate clusters observed on at least seven days with an average nighttime stay duration of no less than four hours. For workplace identification, we define working hours as 08:00–17:00 and select candidate clusters observed on at least four days with an average daytime stay duration of no less than three hours. The centroid of the selected cluster is used as the corresponding coordinate. Overall, the identification rates for home and workplace locations are 87.16% and 83.19%.

*Activity type identification*

Building on identified home and workplace locations, together with the commercial facility dataset, the POI dataset and the land use dataset, we assign each pre- and post-activity to one activity type: *home*, *work*, *shopping*, *public service*, *hospital*, *school*, *leisure*, *transport*, and *other*. We treat *school* and *hospital* as separate activity types rather than grouping them into *public service*, because their visit purpose and temporal patterns are typically distinctive. Activity types are determined by combining facility-specific spatial buffers with a hierarchical priority rule. For each stay activity, we first check whether it falls within the buffer of *shopping* (SC and SS), and then evaluate whether it matches the consumer's *home* or *work* buffer. Next, we test other POI- and land-use-based types. Finally, *transport* is assigned the lowest priority to avoid labeling activities near stations as transport when they overlap with other nearby facilities. Stay activities that do not match any buffer as labeled as *other*.

## 4. Results and interpretation

### *4.1. Identified core consumer groups*

Based on the consumer GPS data processed in Section 3.4.1 and FCPI value introduced in Section 3.4.2, we identified five core consumer groups that exhibit statistically significant preferences across age, gender and facility categories. We extracted 2,078,262 trajectory records for subsequent analyses. The five core consumer groups are follows: (1) G1: prime-age female (30-49) across all facility categories; (2) G2: prime-age male (30-49) across all facility categories; (3) G3: young females (20-29) in SCC, SCO, SCR and SCT; (4) G4: senior females (50-69) in SCC, SCR and SCT; and (5) G5:



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adolescent females (under 19) in SCC, SCR and SCT. Table 1 summarizes their dominant characteristics. G1 and G2 constitute the dominant group. They are high household income (over 7 million JPY) in SCC, SCO, SCR and SCT while low household income (under 3 million JPY) in SS. Within G1, married status is more prevalent in SCC, SCR, SCT and SS, whereas single status is more common in SCO. G2 is predominantly married across all facility categories. G3 is predominantly single in SCC, SCO, SCR and SCT, with low household income (under 3 million JPY) in SCC, SCR and SCT and middle household income (3-5 million JPY) in SCO. G4 is largely associated with high household income (above 7 million JPY) and married status in SCC, SCR and SCT. G5 is predominantly single in SCC, SCR and SCT, with high household income (above 7 million JPY) in SCC and SCT and low household income (under 3 million JPY) in SCR.

### *4.2. Characteristics of consumer behavior*

#### *4.2.1. Temporal characteristics of consumer behavior*

We analyze temporal characteristics using three indicators—visit time, departure time, and shopping duration—to capture when shopping starts, when it ends, and how long it lasts.

Visit time is strongly afternoon-dominated with proportion ranges from 42.91% to 71.38%, while morning visits remain consistently low, with an average proportion below 7% (Figure 6). Visit-time peak emerges differently on day type: non-weekday visits concentrate in the mid-afternoon (15:00–16:00), whereas weekday visits peak later in the late afternoon to early evening (17:00–18:00). Weekdays also exhibit a stronger extension into the night, with a night-visit proportion range 4.47%–25.65%, compared with 2.96%–17.62% on non-weekdays. Notably, G1, G2 and G4 show a relatively higher proportion of morning-shopping, with a proportion range from 6.41% to 8.76%.

Departure time reinforces this day-type contrast and highlights a close time effect on weekdays (Figure 7). Weekday departures cluster toward the close time, with 85% peaking at 19:00. Non-weekday departures shift earlier, with 60% peaking at 15:00–16:00, although 30% still peak at 19:00. Consistently, the night-time departure proportion is higher on weekdays, reaching 18.89% on average versus 14.01% on non-weekdays.

Specifically, SCO is distinctly lunchtime-oriented on weekdays, while other facility categories concentrate later in the day. For visit time, weekday lunchtime happens at SCO reach 36.31% for G1 and 38.37% for G2, both



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Table 1: Core consumer groups and their dominant characteristics

| Group | Facility | Household Income | Marital status |
|---|---|---|---|
| **G1**: aged 30-49 female | SCC | Over 7m (43.86%) | Married (66.14%) |
| | SCO | Over 7m (43.42%) | Single (52.37%) |
| | SCR | Over 7m (40.04%) | Married (67.66%) |
| | SCT | Over 7m (41.85%) | Married (54.28%) |
| | SS | Under 3m (58.23%) | Married (61.06%) |
| **G2**: aged 30-49 male | SCC | Over 7m (50.57%) | Married (61.37%) |
| | SCO | Over 7m (55.12%) | Married (56.41%) |
| | SCR | Over 7m (36.36%) | Married (52.58%) |
| | SCT | Over 7m (49.07%) | Married (56.47%) |
| | SS | Under 3m (55.11%) | Married (54.08%) |
| **G3**: aged 20-29 female | SCC | Under 3m (31.05%) | Single (78.90%) |
| | SCO | 3m-5m (29.38%) | Single (86.96%) |
| | SCR | Under 3m (30.89%) | Single (78.93%) |
| | SCT | Under 3m (32.07%) | Single (86.53%) |
| **G4**: aged 50-69 female | SCC | Over 7m (45.66%) | Married (71.19%) |
| | SCR | Over 7m (44.72%) | Married (72.84%) |
| | SCT | Over 7m (47.31%) | Married (65.66%) |
| **G5**: aged under 19 female | SCC | Over 7m (40.84%) | Single (87.92%) |
| | SCR | Under 3m (47.19%) | Single (89.80%) |
| | SCT | Over 7m (41.65%) | Single (87.62%) |

peaking at 12:00; by contrast, visits to other commercial facility categories for G1 and G2 are concentrated around 17:00–18:00. Departure time exhibits a consistent structure: SCO shows a clear lunchtime peaking, reaching a proportion range from 10.89%-12.50%, while the primary departure peak remains near close time, reaching 13.99% for G1 and 14.45% for G2. For G3, weekday patterns are more continuous from noon into the afternoon: at SCO, the visit-time distribution combines a noon proportion of 27.97% with an afternoon proportion of 59.33%, peaking at 15:00, while peaks shift toward 18:00 in other facility categories, indicating sustained activity into the early evening.

Shopping duration provides complementary evidence that most shopping



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is short (Figure 8). The proportion lasting less than 60 minutes exceeds 50%, ranging from 56.56% to 75.21% on weekdays and from 52.58% to 70.56% on non-weekdays. Short-duration concentration is particularly pronounced in SCT: the proportion of less than 30 minutes is consistently 34.86%–45.75%, peaking at 45.75% for G2 in SCT on weekdays. In SCR, a stronger tendency of less than 30 minutes is also observed for G2, G3 and G5, reaching 37.86% and 42.84% for G2 on non-weekdays and weekdays, 39.82% and 40.57% for G3, and 36.46% and 46.33% for G5, suggesting quicker turnover and more purpose-driven shopping in these contexts.

*4.2.2. Spatial characteristics of consumer behavior*

We summarize spatial characteristics using three indicators—travel distance, leaving distance, and shopping distance—to capture how far consumers travel to reach shopping locations, how far they move after shopping, and how far movements occur within shopping activities.

Travel distance is strongly short-range dominated (Figure 9. Most trips before commercial facilities within 0.1–2 km (58.77%), and the walking and neighborhood scales together account for 66.40%. A secondary range emerges at 3–8 km (16.63%), indicating a non-negligible proportion of cross-area shopping. Facility categories show clear differentiation: SCC, SCR, and SS concentrate more at the neighborhood scale, with proportions of 30.16%, 39.67%, and 32.52%. By contrast, SCO is more strongly associated with the walking scale (36.25%), with 0.1–0.5 km alone accounting for 28.30%, consistent with convenience-oriented shopping. SCT displays a more dispersed profile, with walking, neighborhood, and regional proportions of 30.67%, 25.05%, and 43.66%, implying both nearby and broader-area shopping. For G5, longer travel becomes more visible on weekdays: the 5–8 km band reaches 11.64% in SCT and 8.38% in SCC, suggesting that weekday shopping for G5 can be coupled with longer cross-area mobility.

Leaving distance indicates that post-shopping linkages remain largely local (Figure 10). Neighborhood-scale proportions are 31.91% on weekdays and 31.51% on non-weekdays, and a combination of walking and neighborhood scale account for 67.01% and 65.69%. Metropolitan-scale linkages are marginal, contributing only 0.59% on weekdays and 0.71% on non-weekdays. SCR exhibits a stable neighborhood concentration, with neighborhood-scale proportions ranging from 31.99% to 42.52% on weekdays and from 30.65% to 40.71% on non-weekdays. An exception is G5, for whom the 0.25–0.5 km band reaches 16.29% on weekdays and 27.09% on non-weekdays.



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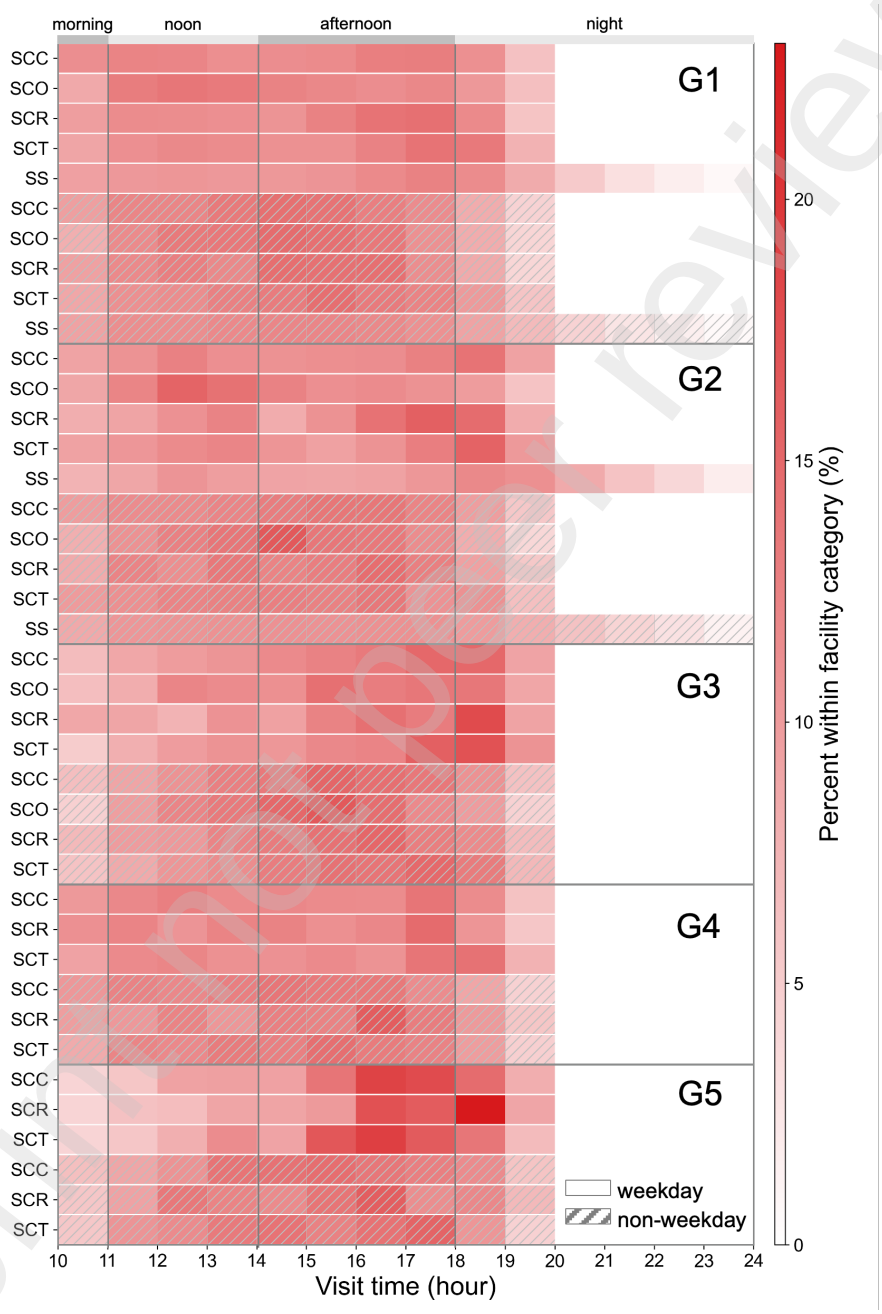



Figure 6: Temporal characteristics of consumer behavior: visit time



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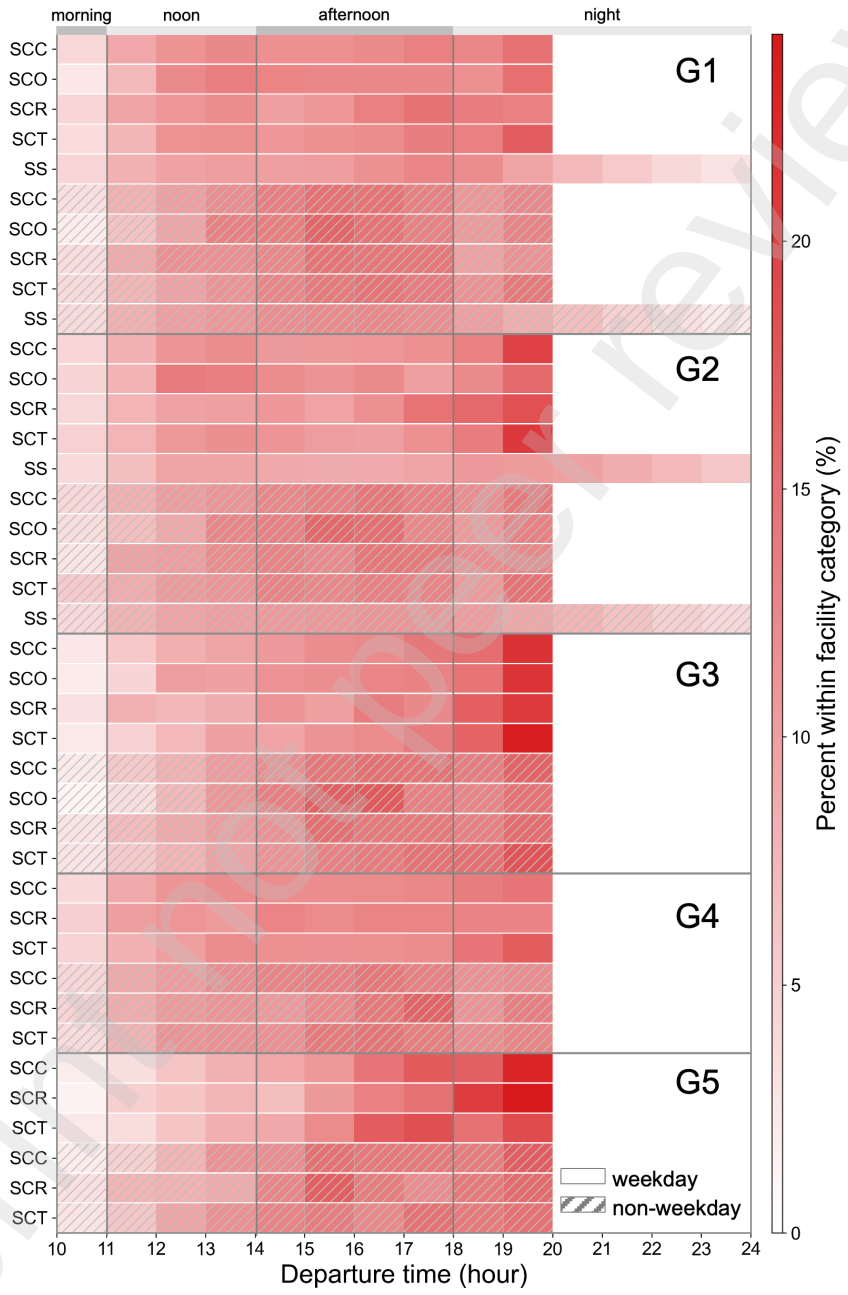



Figure 7: Temporal characteristics of consumer behavior: Departure time



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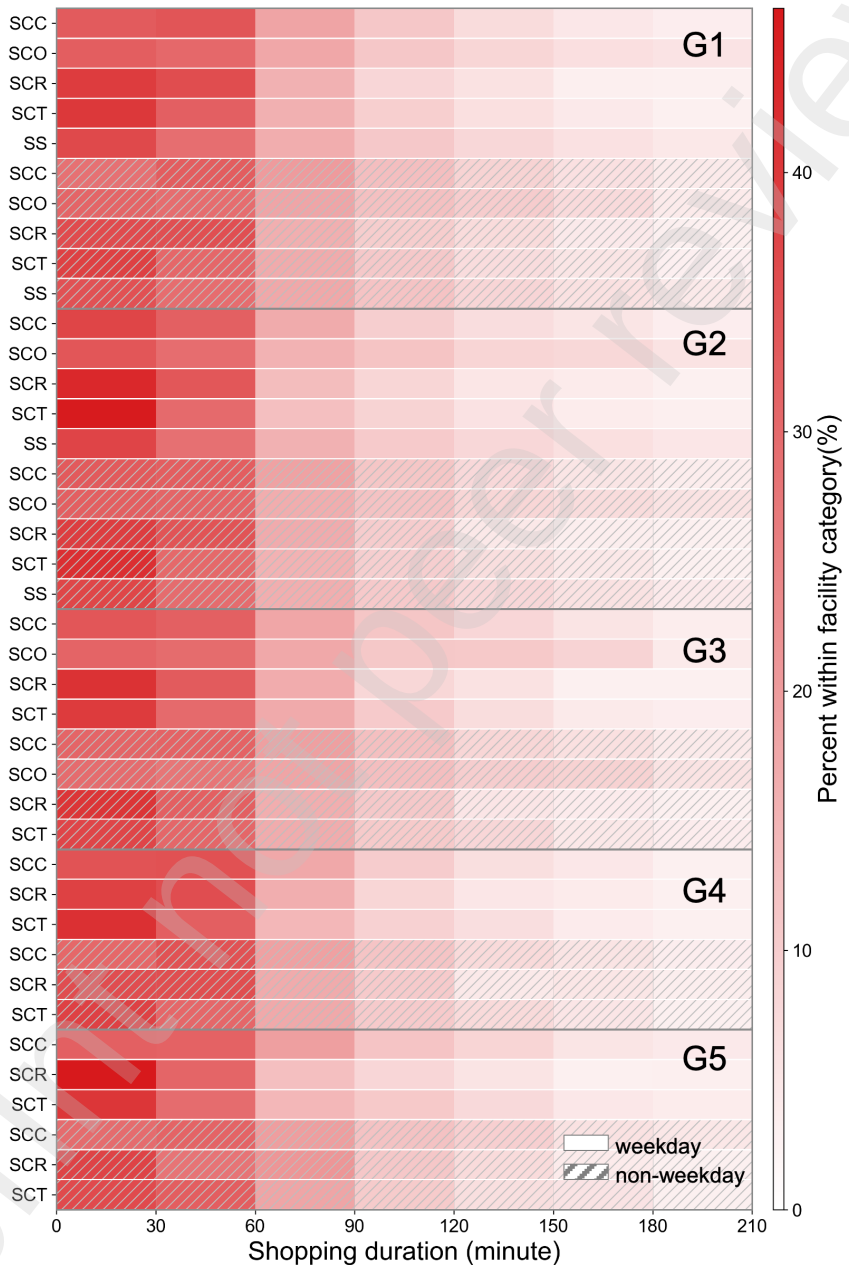



Figure 8: Temporal characteristics of consumer behavior: shopping duration



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Shopping distance reveals obvious differentiation (Figure 11). G1 and G2 exhibit a more dispersed shopping-distance with a proportion range from 27.79%-45.27% among walking, neighborhood, and regional scale. By contrast, G3, G4, and G5 are strongly concentrated within 0.5 km, with proportions of 55.62%, 60.10%, and 59.92%. G4 is particularly localized even within walking range: the proportion of less than 1 km reaches 32.57% on weekdays and 28.48% on non-weekdays, highlighting an exceptionally local shopping intensity.

*4.2.3. Activity sequence characteristics of consumer behavior*

Figures 12 and 13 report the composition of pre-activities and post-activities of *shopping* for each core consumer group. Overall, *shopping* exhibits strong continuity in the activity-sequence. The proportion of pre-shopping is consistently above 60% (60.98%–71.69%), while post-shopping also remains high (58.88%–76.67%), indicating that *shopping* is often embedded in consecutive shopping-related chains. Beyond *shopping*, *home* is the most strongly connected activity type, and post-home is generally higher than pre-home, suggesting a greater tendency to return home after *shopping*. Across all *shopping-home* sequence, the maximum pre-home proportion reaches 19.60% (G2, SCR, non-weekday), whereas the maximum post-home proportion reaches 27.42% (G2, SCR, weekday); notably, within SS, the peak post-home proportion is 27.04% (G1, SS, weekday). *Home* linkage is particularly pronounced in SCR and SS, implying that *shopping* in these facility categories is more frequently embedded in routine *home–shopping–home* activity sequences. A second obvious adjacent activities comprises *public services*, *hospital* and *work*. When combined, their total proportion ranges from 4.74% to 18.95%, indicating that *shopping* is also closely related to daily obligatory activities such as commuting, errands, and medical visits. *Work* is substantially more prominent on weekdays, and pre-work is systematically higher than post-work. The mean proportions of pre-work and post-work on weekdays are 6.75% and 3.09%, compared with 1.78% and 1.25% on non-weekdays. The single highest *work* proportion is 11.21% (G2, SCO, weekday). Finally, while *school* accounts for a small fraction overall, it displays identifiable peaks in specific cases. For instance, post-school becomes relatively more visible in SCR for G5 and G4 on non-weekdays, reaching 0.74% and 0.52%. In contrast, *leisure*, *transport* and *other* remain marginal no more than 3%, reinforcing that the dominant activity-sequence structure is primarily shaped by *shopping–home–(public-service/hospital/work)*.



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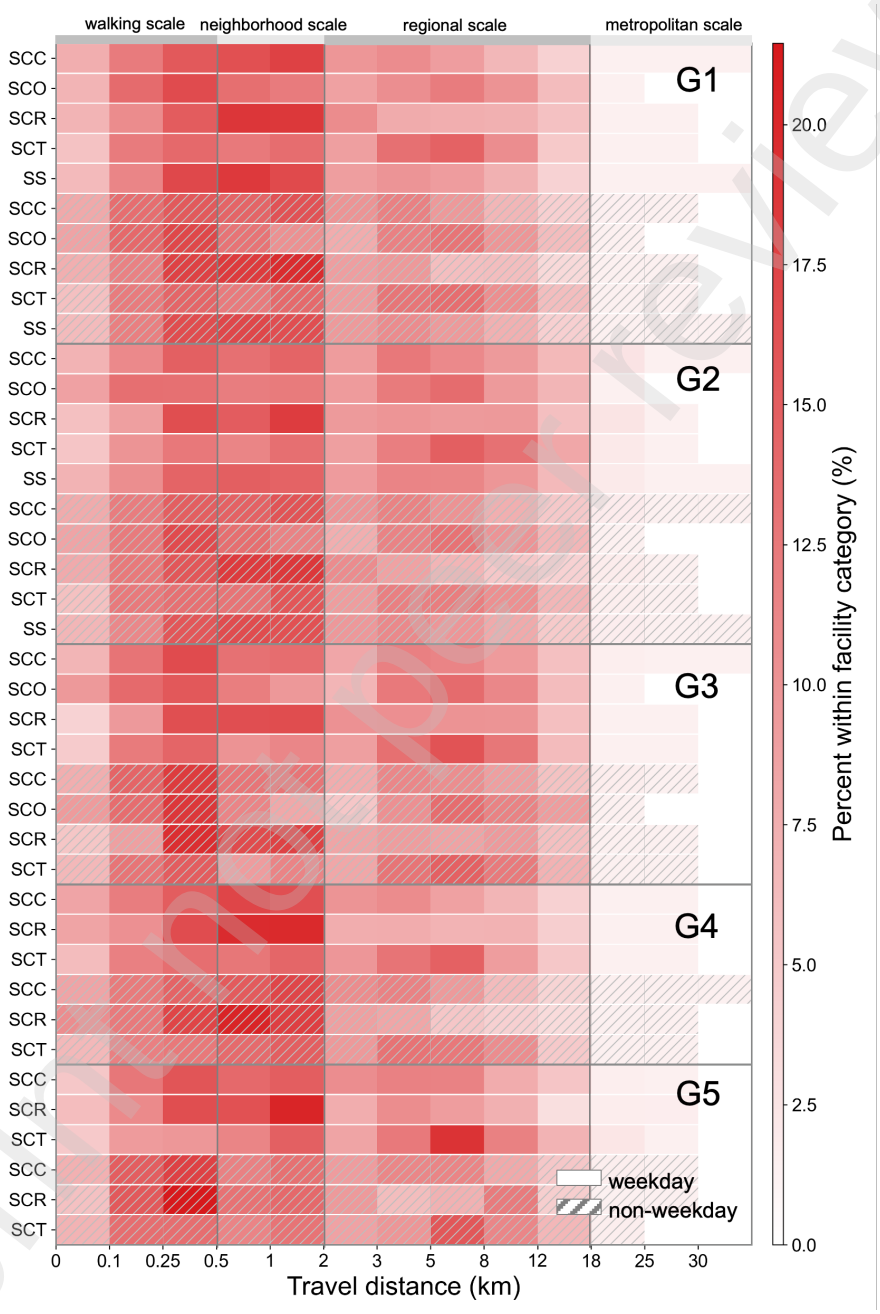



Figure 9: Spatial characteristics of consumer behavior: travel distance



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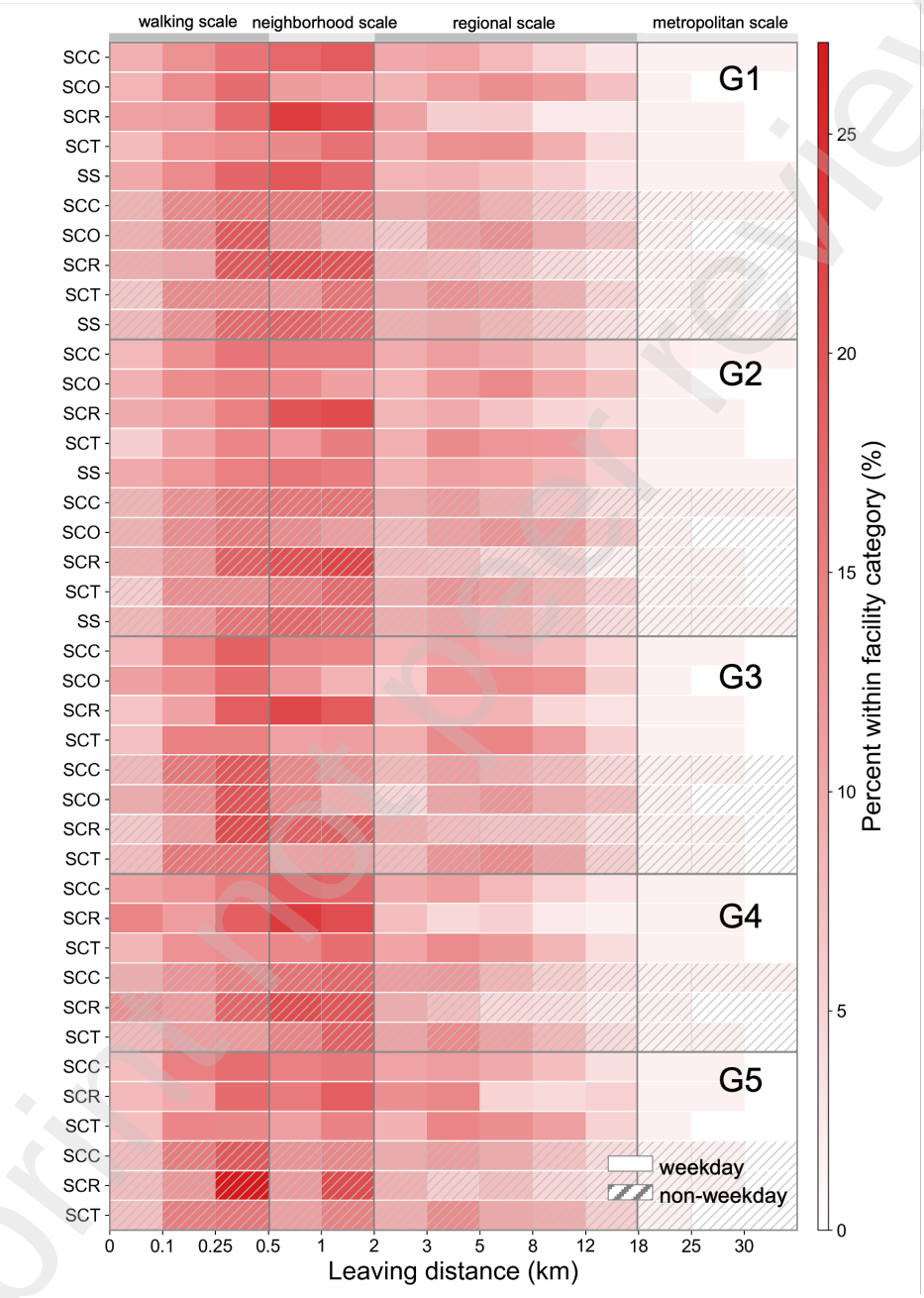



Figure 10: Spatial characteristics of consumer behavior: leaving distance



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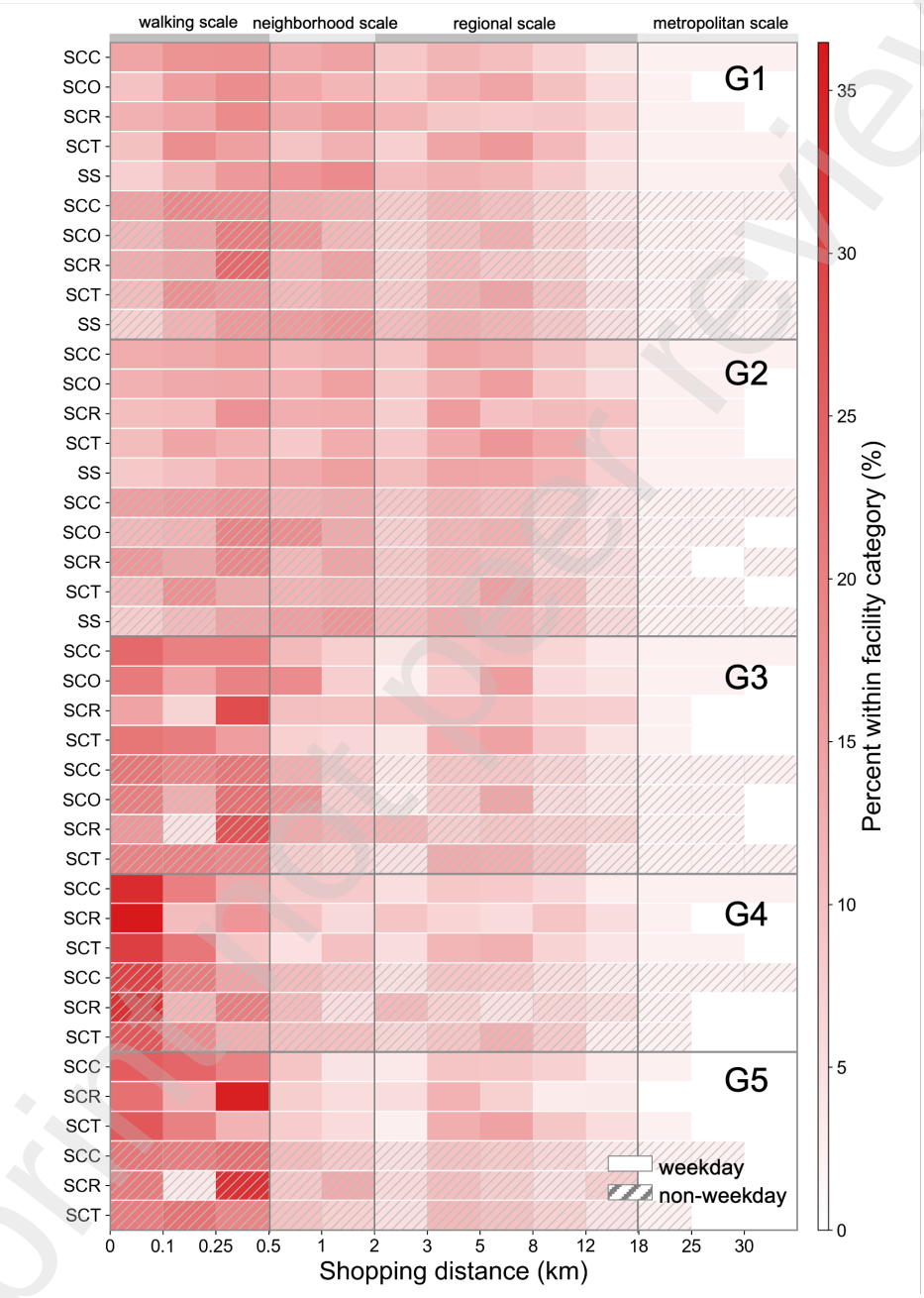



Figure 11: Spatial characteristics of consumer behavior: shopping distance



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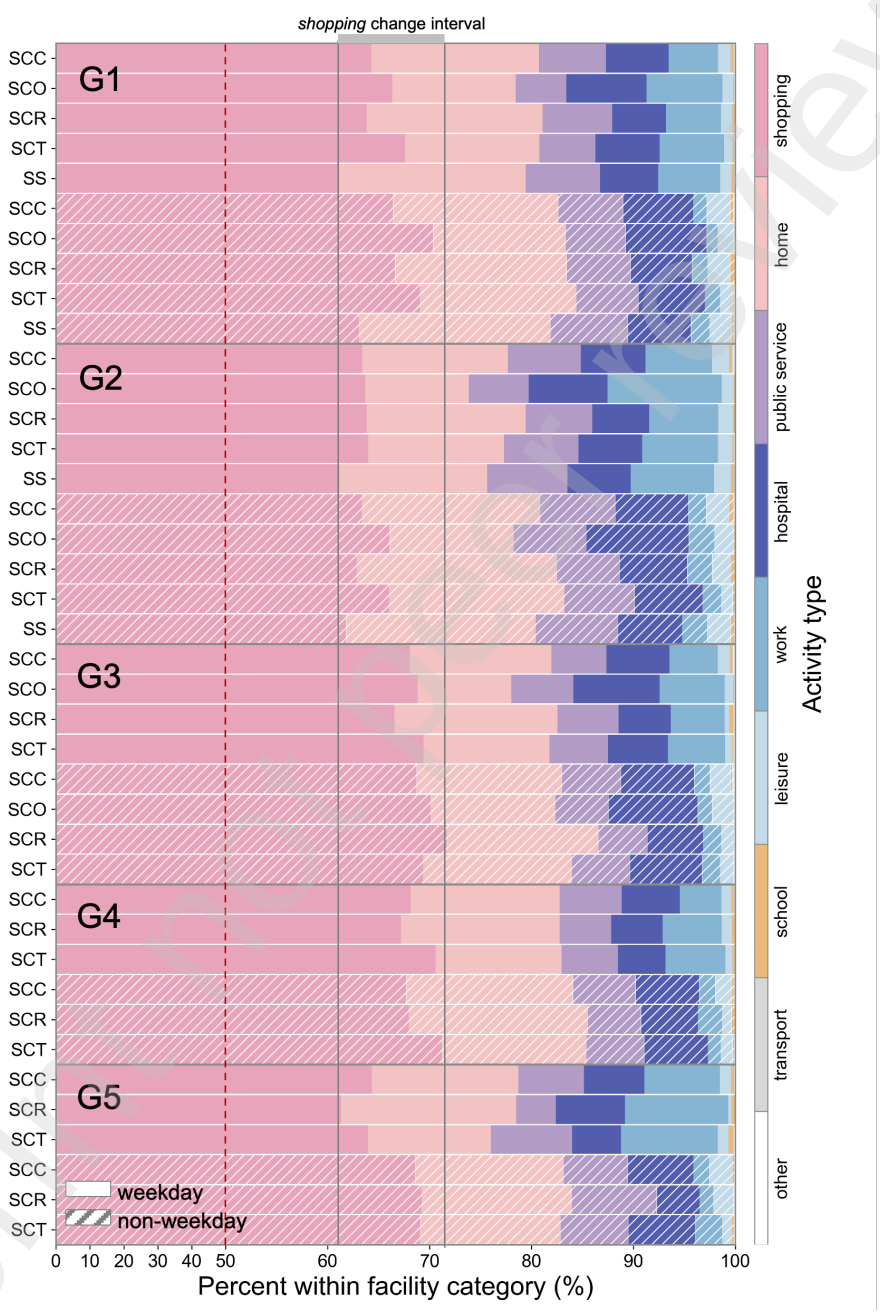



Figure 12: Activity sequence characteristics of consumer behavior: pre-activity



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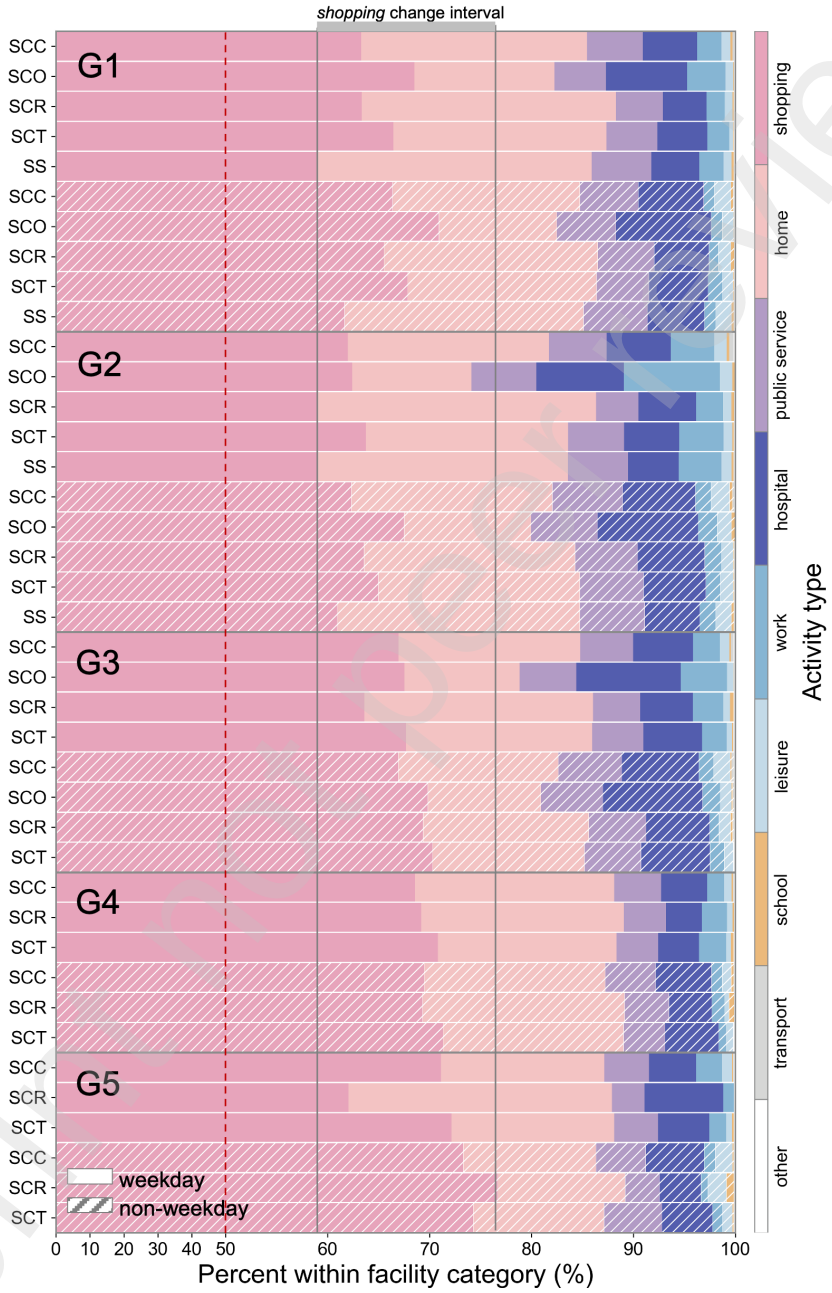



Figure 13: Activity sequence characteristics of consumer behavior: post-activity



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### *4.3. Consumer behavior patterns through temporal, spatial and activity sequence characteristics*

We organized four consumer behavior patterns. (1) Routine shopping pattern (Figure 14): *shopping* exhibits strong sequential continuity, and most happens from daytime to the afternoon within a 60-minute activity time and a 500-meter activity range. This pattern is primarily organized at walking- and neighborhood-scale distances, with occasional extensions to the regional scale. *shopping* is also stably connected to *home* and, secondarily, to obligatory activities such as *public service*, *hospital*, and *work*. (2) Work-embedded shopping pattern (Figure 15): building on the characteristics of routine shopping pattern, this pattern shifts shopping activities into weekday work rhythms for G1, G2 and G3 at SCO, where *shopping* occurs at the walking scale within a 60-minute activity time and a 500-meter activity range, and is constrained by *work*-related schedules. Two subtypes are evident: lunchtime (lunch-break) *shopping* is more prominent in SCO, whereas dinner-time (after-work) *shopping* is more common in non-SCO facility categories. (3) Neighborhood shopping pattern (Figure 16): building on the characteristics of routine shopping pattern, this pattern highlights community-oriented needs for G4 at SCC, SCR and SCT, where *shopping* occurs at neighborhood scale within 60 minutes and a 100-meter activity range and is concentrated during daytime. Notably, *school* is relatively frequent on non-weekdays following *shopping* at SCR. (4) School-anchored shopping pattern (Figure 17): building on the characteristics of routine shopping pattern, this pattern has *shopping* in after-school time of weekdays for G5 at SCC, SCR and SCT, where *shopping* typically occurs at walking or neighborhood scale within a 60-minute activity time and a 500-meter activity range. Notably, *work* more often appears before *shopping* on weekdays, and *school* is relatively frequent on non-weekdays following *shopping* at SCR.

## 5. Discussion and implications

### *5.1. Exploring consumer behavior patterns in commercial space from the perspective of mobility*

#### *5.1.1. Identifying interpretable core consumer groups through shopping preference intensity*

This study identifies core consumer groups using the FCPI value, narrowing the analytical focus from the broader set of visitors to core consumer



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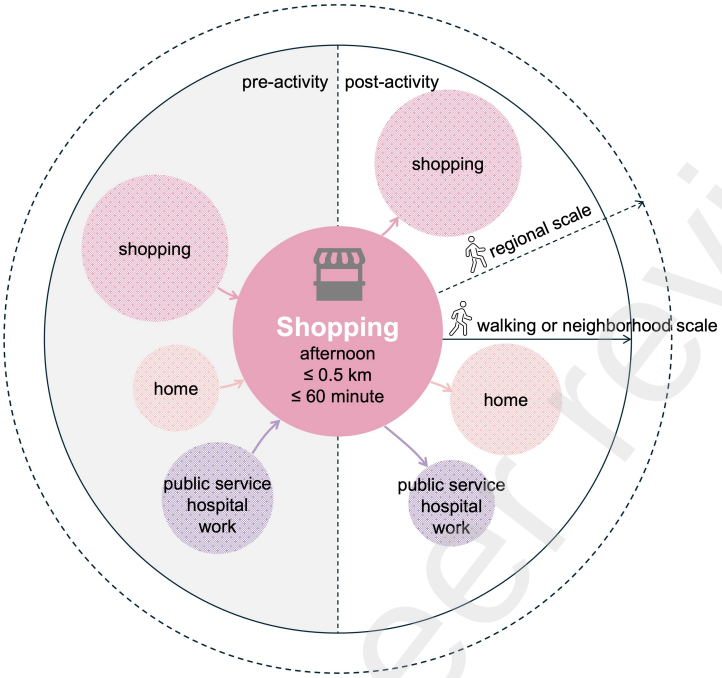



Figure 14: Routine shopping pattern

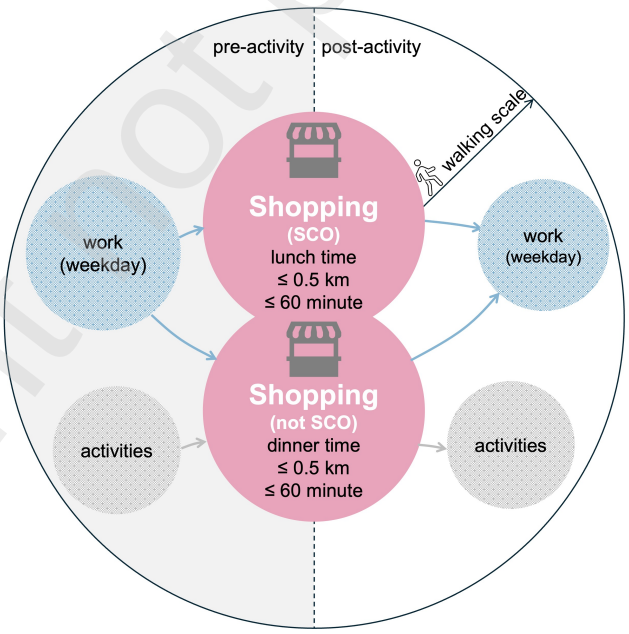



Figure 15: Work-embedded shopping pattern: G1, G2 and G3



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Figure 16: Neighborhood shopping pattern: G4

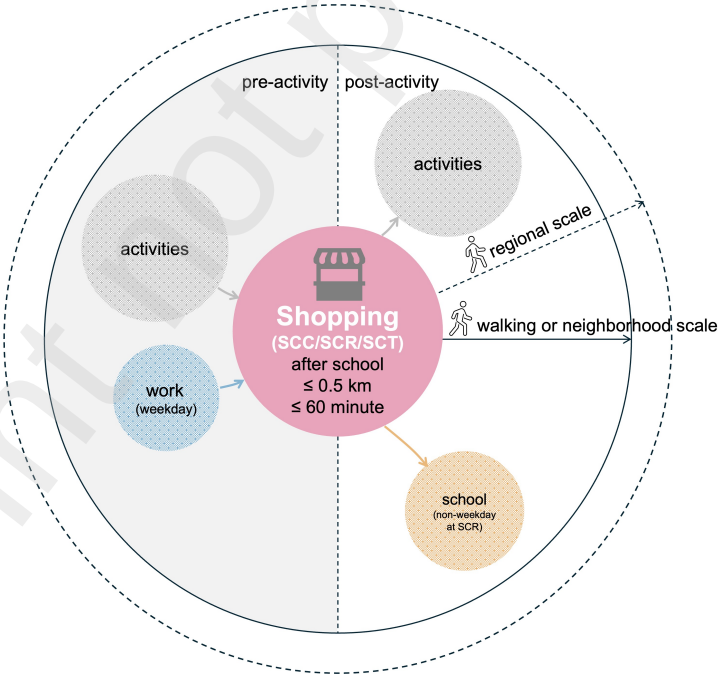



Figure 17: School-anchored shopping pattern: G5



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groups that exhibit a strong preference for specific commercial spaces. This approach enables subsequent consumer behavior analysis to better reflect underlying attraction mechanisms: core consumer groups not only visit frequently, but also demonstrate stable shopping preferences, such that their behavior patterns are more representative of the key demands and usage contexts served by each facility category. Based on gender, age, and facility categories, we identify the following five core consumer groups.

**G1: prime-aged women (30–49) balancing career and family responsibilities.** This group typically combines relatively stable purchase power with diverse household-related demands, spanning routine procurement, child-related spending, and family-oriented activities.

**G2: prime-aged men (30–49) with an efficiency- and productivity-oriented profile.** Their shopping tends to be purposeful and time-constrained, often aligned with work rhythms and commuting routes, reflecting a more embedded organization within daily routines.

**G3: young women (20–29) characterized by independence and experience-oriented consumption.** Quick transition from study to employment with more variable schedules, they are more likely to combine functional purchases with experience-oriented shopping.

**G4: senior women (50–69) as active participants in the silver economy.** As a result of social development in Japan, this group with a relatively good health, flexible time, and economic capacity. They show active and diversified consumption, consistent with "active aging", and represents a key driver of the silver economy.

**G5: adolescent females (under 19) oriented toward school-linked and peer social activities.** Constrained by school schedules, their shopping often accompanies peer interaction, after-school stopping, and small everyday purchases in exploratory leisure contexts.

*5.1.2. Constructing a comparable analysis framework across five commercial facility categories*

We also account for differences across facility categories (SCC, SCO, SCR, SCT, and SS) in our analysis. It provides structured and interpretable classification of commercial spaces, rather than treating commercial places as a single homogeneous category. The five categories reflect systematic differences in functional composition, tenant mix, typical service types, and daily usage contexts, ranging from nearby, convenience-oriented categories to more multi-functional categories.



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#### 5.1.3. Understanding of consumer behavior patterns through temporal, spatial and activity-sequence dimensions

The temporal dimension reveals daily rhythms cross weekdays and non-weekdays. The spatial dimension captures variation in travel distance, service range and influencing range. The activity-sequence dimension situates shopping within daily activity chains and directly observes how shopping connects with home, work, and obligatory activities such as public services, education, and healthcare, thereby providing a basis for interpreting shopping motives and task settings. Based on these three-dimensional characteristics, we identify four consumer behavior patterns to explain how different consumers organize shopping mobility across commercial spaces in their daily life: routine shopping, work-embedded shopping, neighborhood-anchored shopping, and school-linked shopping.

### 5.2. Implications for commercial development

Four consumer behavior patterns help translate consumer demand into targeted practical implications for commercial development and urban support systems.

#### 5.2.1. Supporting multi-purpose routine shopping chain within neighborhood ranges

Routine shopping pattern represents the most typical consumer behavior. First, from the perspective of shopping continuity, it implies a clustering effect of commercial spaces. When shopping facilities are more spatially concentrated and routes between them are smoother, consumers can complete multiple shopping within a small area, reducing cross-area travel costs. Second, from the perspective of 2 km activity range, mobility convenience is dominant, which can shape whether shopping can occur smoothly, such as continuous pedestrian networks, weather-protected walkways, safe crossings, and barrier-free environments. Third, from the perspective of multi-purpose shopping, many facilities function as everyday service nodes rather than purely retail destinations. This supports tighter integration of commercial facilities with clinics, pharmacies, administrative services, and other errand-oriented functions, enabling consumers to accomplish multiple tasks within a single outing. Notably, in some settings this pattern coexists with married and relatively higher household-income groups, implying that efficiency-driven errands can coexist with household consumption oriented toward quality and reliability. Accordingly, the priority is not only proximity,



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but also the provision of everyday services that are trustworthy, stable, and efficient. Finally, from the perspective of operational strategies, the higher tendency to return home after shopping and the pronounced closing-time effect suggest that facilities should strengthen their role as the last activity before going home. Fast checkout, pre-order and pick-up services, parcel lockers, ready-to-eat and semi-prepared dinner options, and clearly designed convenience-oriented routes can reduce congestion and improve service efficiency.

#### *5.2.2. Fostering convenient, socially interactive, and diverse commercial spaces for workers*

Work-embedded shopping pattern mainly support for workers. First, from the perspective of visit time and travel distance, shopping around lunch and dinner hours highlights the importance of micro-scale linkages between office areas and nearby commercial facilities. Whether lunchtime shopping can occur often depends on the continuity of the shortest walking route, the availability of weather-protected paths on rainy days, waiting time at crossings, and congestion near entrances. After-work shopping similarly relies on seamless connections to commuting routes and public transit. Notably, SCOs show a service gap for those shopping after work. In a high-density city such as Tokyo, late-night overtime is not uncommon, and relevant facilities should provide more readily accessible replenishment and services for these workers. In addition, the tendency for after-work workers to gather in shopping-dense districts for socializing or drinking implies that extending selected functions beyond 21:00 may better match actual after-work rhythms. Entrance placement, internal circulation, and sign should be coordinated with major commuting paths to minimize travel cost. Second, from the perspective of shopping duration and distance, beyond quick dining and resting services, SCOs can incorporate coffee-style public spaces to support flexible stays, such as informal work, quick meetings, and face-to-face communications. Prior studies suggest a positive association between the number of coffee shop and applications for patents [**?** **?** ], which provides additional rationale for integrating coffee-style public workspaces into office-oriented commercial facilities, with potential benefits for light-weight productivity and social interaction. Third, from the perspective of worker attributes, service provision should be structured as a stratified structure. Married and high income workers tend to prioritize time-saving and high-quality products, whereas younger and single groups may require more flexible pricing and more relaxed consumption



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settings.

*5.2.3. Building neighborhood commercial environment for seniors*

Neighborhood shopping pattern mainly shows a relatively balanced distribution across the daytime. First, from a planning perspective, this suggests that facilities can attract them to visit at earlier hours, thereby relieving concentration pressure in the afternoon and early evening. Second, in terms of shopping distance and travel distance, a task-oriented shopping at nearby facilities ask managers can use compact entrance to reduce shopping time. And services such as pre-order pickup, assisted packing, and short-distance delivery are better embedded as part of the community commercial system, supporting better service rather than relying on low-price competition. Third, in terms of activity sequences, G4 at SCR shows a small but identifiable linkage to school-related activities on non-weekdays after shopping, which may reflect pick-up children for after-school lessons or participation in community activities at school. Accordingly, neighborhood layout along the shopping-school chain can be improved to better support G4's neighborhood mobility.

*5.2.4. Designing after-school shopping accessibility*

School-anchored shopping pattern mainly occurs after school, with a more dispersed travel distance at SCT and a more concentrated travel distance at SCR. First, from the perspective of urban planning and transport systems, connectivity to commercial facilities should not rely solely on rail, but should be strengthened by walking and cycling networks, bus feeders, and rail transfers, while provide public spaces to support short stays and peer interaction. Meanwhile, connectivity to home should prioritize convenience and safety, reducing detours and waiting time so that shopping activities can be completed efficiently. Second, from an activity-sequence perspective, transit-oriented commercial facilities can support student discounts and short-duration events to match after part-time students and students who tend to visit school on non-weekdays; and residential-oriented commercial facilities should emphasize high-efficiency services before going home.

*5.3. Limitation*

First, because Tokyo is highly functionally dense and our activity labeling relies only on the MLIT POI and land use dataset and the CSIS POI dataset, both of which have limited category granularity and incomplete coverage, and because we lack external ground-truth validation, the POI-based



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activity types may be insufficiently fine-grained and may affect the precision and completeness of the activity-sequence analysis. Second, we did not further divide multiple shopping activities made by the same consumer within a single day, which may merge repeated shopping into a single shopping activities thus affect the characterization of long-duration consumers. Third, this study focused on residents of the 23 wards of Tokyo, without consideration of tourism.

## 6. Conclusion

This study focuses on the 23 wards of Tokyo and adopts a dynamic mobility perspective to integrate temporal, spatial and activity-sequence dimensions into an analytical framework for understanding consumer behavior patterns within different commercial spaces. By identifying five core consumer groups characterized by high shopping preference intensity, we summarize four consumer behavior patterns across weekdays and non-weekdays. The findings reveal that consumer behavior patterns exhibit both commonalities and heterogeneity. These behaviors manifest in four distinct patterns: (1) routine shopping, which exhibits strong sequential continuity, is mostly anchored by home and other obligatory activities, and predominantly occurs in the afternoon within a 60-minute duration and a 500-meter spatial range, governed by walking and neighborhood scales; (2) work-embedded shopping, shaped by lunch and after-work schedules; (3) neighborhood shopping, characterized by a highly localized spatial range; and (4) school-anchored shopping, organized around after-school mobility. In addition, this study discusses four practical and micro-scale implications for urban commercial development: (1) supporting multi-purpose routine shopping chain within neighborhood ranges, (2) fostering convenient, socially interactive, and diverse commercial spaces for workers, (3) building neighborhood commercial environment for seniors, and (4) designing after-school shopping accessibility. Collectively, these measures facilitate a closer alignment between commercial operations and observed consumer behavior patterns, thereby strengthening the adaptive capacity and resilience of urban commercial systems.



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